foundation capital

# Physical Intelligence

CVC Deliverable Spring 2026

# MARKET OVERVIEW

The physical AI stack is maturing unevenly across five layers, creating pockets of structural underinvestment where early-stage startups can build defensible positions before incumbents consolidate

## Overview

- Durable positioning requires integration across layers; companies that collect data through deployed fleets outperform single-layer specialists
- The most defensible startups generate proprietary data as a byproduct of commercial deployment
- Over 60% of robotics development relies on open-source ROS2, yet no implementation meets current industrial safety certification requirements
- Generalist models reach roughly 60% capability in robotics vs. 80% for LLMs, requiring verticalized tuning per application domain
- Hardware pricing power is eroding as Chinese OEMs close the quality gap, pushing Western startups toward compliance-heavy verticals

## Physical Intelligence Stack

**Hardware**
Actuators, sensors, edge compute, power systems, and precision components

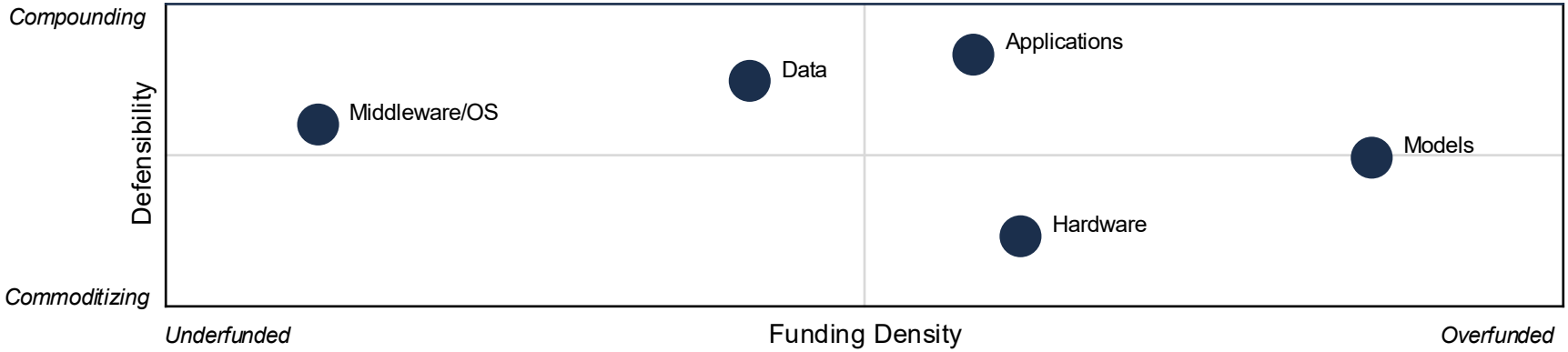
**Data**
Real-world collection, synthetic generation, annotation, and curation pipelines

**Models**
Foundation models, world models, and vertical fine-tuning for perception & control

**Middleware/OS**
Fleet orchestration, safety-certified runtime, and simulation-to-deployment tooling

**Applications**
Field service and primary care robotics deployed in real-world environments

## Areas for Specialization




Sources: [1] a16z [2] ArXiv [3] Berkeley [4] Deloitte [5] Epoch [6] Macenski et al., Science Robotics [7] NVIDIA

# INDUSTRY HEADWINDS

Bottlenecks in computational and hardware efficiency, coupled with legacy integration friction, delay widespread physical AI implementation, particularly for retail consumers

- **Compute is unable to be used effectively due to the lack of high-quality robotics data**
  - Leading robot manipulation models use ~1% of the computational power for training compared to leading models in other domains, like LLMs, which is a result of a scarcity in training data
  - To train foundational LLMs (ex. Llama 3), you would need about “100,000 years” of training data in the form of text, images, videos, haptic, etc. (at least 900PB of data); the largest robotics dataset (Open-X Embodiment) contains <10k GB of data, and there are far less robotics vs. text datasets
- **Integration into existing infrastructure is a significant hidden cost for enterprise deployment**
  - According to Deloitte, 35% of industry leaders attempting to integrate embodied AI cite integration as “the greatest barrier to adopting physical AI in your organization” due to high hidden costs
  - Licensing fees (the “visible cost”) of enterprise use for software AI is only 20-30% of the total cost of the product, with hidden costs ranging from data preparation, integration, and model validation
  - Embodied AI has a higher average overall OpEx cost compared to software AI (~$8000 vs. <$5000), further lowering the percentage of total cost visible to potential consumers
- **Hardware technology limits processing speeds and mobility of battery-dependent robots**
  - Latency between robot input and responses presents a need for edge AI computing hardware, which depends on having equally fast high-bandwidth memory (HBM); though peak compute speeds for AI hardware has increased by 60,000x over the past 20 years, interconnect bandwidth has increased by only 30x in the same time period, meaning newer compute hardware cannot efficiently use memory
  - Battery technology is limited by low energy density—mobile robots not able to be connected to power at all times (ex. humanoids, delivery robots, etc.) are usually made up of ~20% battery by mass, forcing a reduction in the mass of other hardware included like processors, sensors, and actuators
  - Higher density battery technologies (ex. zinc-air batteries) suffer from fewer or no recharge cycles and increased weight, so they lessen the lifetime and make movement more difficult for mobile robots

# INDUSTRY TAILWINDS

Active deployment of commercial robot fleets, coupled with expanding synthetic data infra, is widening bottlenecks in robotics training and manufacturing that will improve timeframes and product performance

- **Declining costs of core robotics hardware is unlocking mass deployment**
  - Significant advancements in chip, LiDAR, and camera module manufacturing have helped pull robotics into dynamic, unstructured settings such as warehouses, hospitals, and retail
    - LiDAR costs have fallen by upwards of 90% in the last decade
- **A powerful flywheel in data and manufacturing is emerging as robotic systems are deployed**
  - Early movers deploy fleets that both serve their clients' needs and generate the proprietary data needed to improve their models. As deployment scales, so does data quality, leading to a powerful self-reinforcing advantage that compounds over time
    - Companies such as Waymo find their edge in its scale, talent, and data flywheel from its fleet of 3000+ vehicles, which all serve as both products and data collectors, with 200 million miles of fully autonomous, real-world driving experience as of February 2026
  - Robotic systems are maturing as production becomes both proven and more cost-effective, and robotic system implementation becomes more financially viable for firms
    - Since 2019, average industrial robot payback times have dropped from 5 years to 1.3 years
- **Scaling synthetic data and foundation models are improving early-stage development**
  - World Foundation Models and other sources of synthetic data are dramatically lowering the costs and timeframes associated with robotics development, leading to faster iteration cycles before real-world deployment and testing
  - NVIDIA's Omniverse and Cosmos world foundation models have shown true dominance in the robotics space, with millions of downloads by developers across manufacturing, the automotive industry, and humanoid robotics. The scale of these datasets has democratized early-stage training
  - Smaller companies such as Parallel Domain use synthetic data to improve robotics development by providing developers with rapidly deployable virtual training environments

Sources: [1] AI Vision [2] Data Bricks [2] DigitalEngineering [3] NBC [4] McKinsey [5] Parallel Domain [6] Scanford

# HARDWARE LAYER

Defensibility is shifting from the consolidated sectors of precision mechanics, manufacturing, and edge compute, to software defined intelligence in high compliance verticals which circumvent physical constraints

## Overview

- This layer contains atoms (actuators, sensors, power systems) and bits (edge-compute and communication frameworks) required to translate digital intelligence into physical work
- In the deployment of Physical AI, the primary constraints are physical factors like torque density of actuators, latency, and precision of feedback loops
- The layer is split into the US-led innovation in high-end silicon (bits), while China dominates in mass production and industrial ecosystems (atoms)
  - China maintains a 30-50% cost advantage on precision harmonic reducers
- This commoditization of manufacturing shifts the defensibility to the software behind the robots
- The most consolidated domains are
  - Edge Compute (NVIDIA: Jetson Thor chip)
  - Memory (South Korea: Samsung and SKH)
  - Industrial Robotics OEMs (FANUC, ABB, Yaskawa, and Kuka)
  - High Precision Reducers (Japanese giants HDSI and Nabtesco)
- China and Japan's dominance on the “atom” layer, and Western consolidation of edge compute (due to NVIDIA's proprietary platform CUDA), forces new players to focus on high-compliance verticals like space or defense. Additionally, these sectors benefit from advances in simulation due to high R&D costs

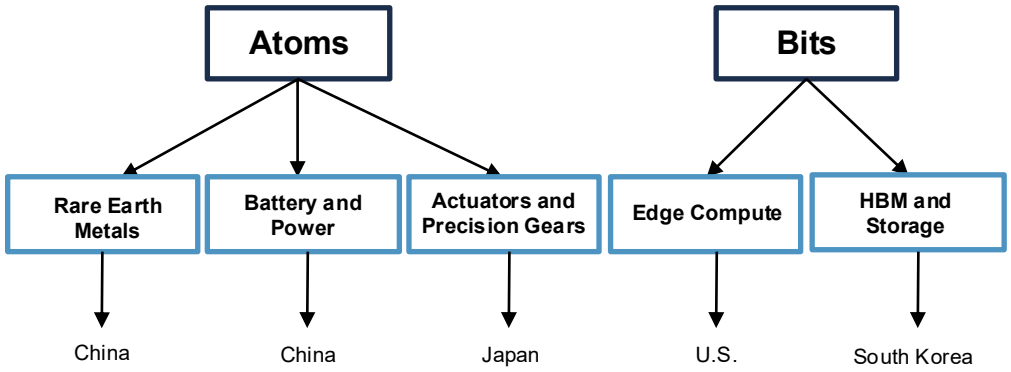
## Primary Headwind

- **Surge in Component Crowding** has led to the robotics industry having to compete for specialized hardware (high-bandwidth memory, power management ICs, and thermal solutions) against better capitalized sectors like data centers and EVs

## Primary Tailwind

- **Reduction in R&D Costs** due to increase in Sim-to-Real accuracy. Emerging parties no longer have to worry about precise “atom” investment. Additionally, high-end hardware prices are falling (Collaborative Robots are now available for <$30k) allowing SMEs to prove reliability for cheaper

## Layer Breakdown



# THE DATA LAYER

Growth in the robotics data layer is driven by the need for high-quality labeled datasets and training environments, especially for edge cases and accuracy-critical applications

## Overview

- High-quality data is gathered using a variety of sources, including deployed robots with sensors, human movements for imitation, teleoperator-controlled machines, reinforcement learning environments, and synthetically generated data methodologies
- Synthetic data and simulated environments serve as a strong scaling tool for early-stage companies, but fall short in capturing the complexity and edge cases of more chaotic real-life application environments
- Robotics data annotation needs to capture spatial relationships and physical interactions in a unified and consistent way, making it far more technically demanding than traditional computer vision tasks

## Synthetic Data And Foundation Models

- Perception data is increasingly commoditized by foundation models, spanning synthetic data generation and RL environments, but is structurally limited in late-stage training due to difficulties in modeling real-life interactions and edge cases, common to AVs and humanoid robotics
- While annotation of sensor streams, video, and point clouds remains essential, traditional labeling faces structural constraints, and in the near term, foundation models are increasingly automating much of that work
- The cutting edge of model training has shifted toward real-world operational data, including physical data collection from robotic systems deployed in the real world
- As synthetic data matures, the gap between real and simulated will shrink. Regardless, real-world fleet data remains hardest to replicate

## Data Stack

### Synthetic Providers

NVIDIA

parallel domain

### Labeling

scale

iMerit

## The Real-World Data Moat

- Real-world data, especially for robotic manipulation and dexterity (Figure AI, Sensei), remains difficult to simulate and is unlikely to be commoditized, as high-quality real-world data gathered by operators cannot be generalized across applications as effectively as synthetic data, and operators prefer to keep proprietary data in-house as it becomes an edge over competitors
- Operators with active fleets hold a significant structural data advantage as product deployments become data collection, which continuously log sensor streams, edge cases, and environmental interactions across real-world conditions, creating a real-world data flywheel that cannot be replicated by synthetic platforms or foundation models

Sources: [1] BlackGibbon, [2] DigitalEngineering, [3] Data Bricks, [4] NVIDIA, [5] Scanford

# THE DATA LAYER

Synthetic data production is approaching a limit in its promised RL training efficiency gains, shifting the layer focus to the metadata surrounding datasets to maximize training efficacy

## Viability of Synthetic Data

- Training via simulation has sped up dramatically; NVIDIA Isaac allows for RL training without a CPU, reducing training time by 2-3 orders of magnitude
- Weaknesses arise from the complexity in contact modeling and sensor noise replication (ex. material deformation, friction coefficients, and sensor white noise are all difficult to estimate with high accuracy)
- While simulation delivers a 70% reduction in training time, applications requiring high precision (ex. medicine, dangerous chemicals, etc.) still need to integrate real-world data for the foreseeable future

## Data Layer Categories

**Teleoperation**: manual collection

**Synthetic Data**: simulation models

**Data Labeling**: manual/automated annotation tools

**Data Curation**: filtering training datasets

## Areas for Specialization

- **Labeling:** Multimodal robotics data has long relied on manual annotation; automated labeling platforms still rely on human annotation to train their models on (ex. ARES from a16z, autoLabel from Samsung)
- **Curation**: Current models seem to favor data quality and diversity over quantity: “a model trained with curated, high-quality demonstrations achieves 40% better throughput despite being trained with ⅓ less data” (Figure AI when training custom VLA model)
- **Specialized Datasets:** Many needed data types are sparse (ex. Outdoor seasonal data from ROVER, warehouse robotics datasets needed by Otto Motors)

## Market Trends

**Major acceleration in robotics dataset creation**: 1K in 2024 to >25K in 2025

**Sim creation more accessible**: Spatial cameras, collection robots, and closed-loop Real2Sim

**RL training rapidly sped up**: Unified GPU build & parallel execution

**Dataset curation over dataset size:** Models perform better given diverse and full-coverage datasets

Sources: [1] a16z [2] AI World [3] EdgeAI [4] Inbolt [5] InOrbit [6] Figure AI [7] NVIDIA [8] Precedence [9] Samsung SDS [10] Scale AI [11] XGRIDS

# THE MODEL LAYER

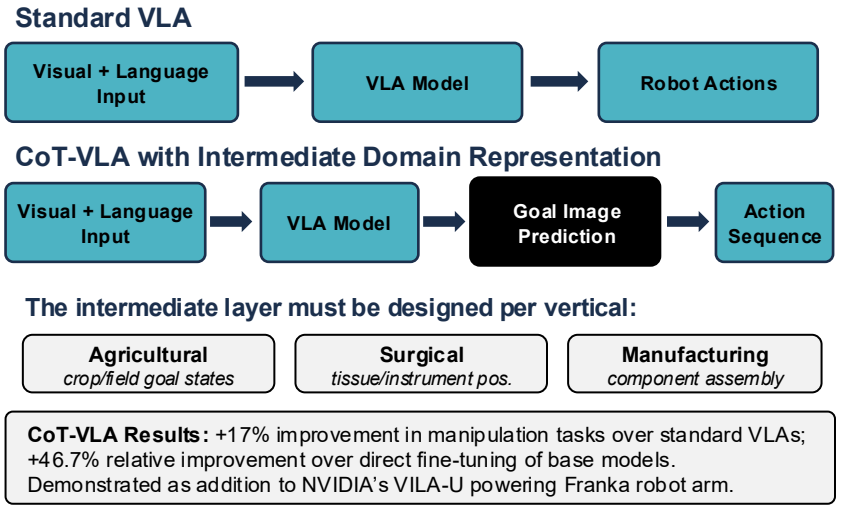
With frontier model training still concentrated in leading players of compute and talent, the pre-seed/seed opportunity lie in architecture innovation, domain-specific intermediate layers, and applied development

## Overview

- The model layer includes both foundation models for learning general tasks across perception, planning, and control, along with world models for understanding real-world physics and generating synthetic training environments for robots
- Open-source foundation models are closing the capability gap, shifting importance to developing fine-tuning and deployment layers
  - COSMOS Policy (NVIDIA) and OpenVLA-OFT are achieving similar success rates of 90% to closed-source Pi05 on LIBERO benchmark
- Leading foundation models and world models are both dominated by late-stage players, leaving no clear entry point for seed-stage startups to build generalized robotics models
  - Frontier model training requires ~$50M-$200M in compute while top researchers command $20M in compensation
  - NVIDIA and Google DeepMind lead adoption and distribution of world models
- Robotics companies treat model layer as something to procure, with top in-house models being fine-tuned versions of general foundation models, commoditizing frontier models while value accrues across fine-tuning and deployment

## Architecture Innovation

- Startups with deep domain insights create value by enabling model architecture innovation for general model providers, optimizing these robotics foundational models for specific application verticals without adding complexity through fine-tuning
- CoT-VLA model predicts what environment should look like after completed goals and generate action sequence to reach them, while GROOT N1, Physical Intelligence, and Gemini Robotics 1.5 have all begun to add intermediate layers to their model architectures, creating demand for implemented domain representations across robotics verticals




Sources: [1] Air Street Press [2] CoT-VLA [3] Epoch AI [4] Gemini Robotics [5] NVIDIA Research [6] OpenVLA [7] Reuters

# THE MIDDLEWARE/OS LAYER

The robotics market needs a hardened ROS2 model with improved security, safety, and real-time guarantee. Owning this position creates recurring revenue from companies that cannot run on standard ROS2

## Overview

- Robot middleware bridges a robot's hardware and software, facilitating communication and integration between components. Its effectiveness depends on interoperability, scalability, real-time processing, modularity, and fault tolerance. Common frameworks include ROS and OROCOS
- This layer has two components: an operating system (OS) and a middleware layer on top that handles inter-process communication and abstracts away differences in hardware systems

## Key Challenges with ROS2

- ROS2's default scheduler cannot guarantee safety-critical tasks will meet timing deadlines. Formal verification is theoretically viable, but no open-source implementation is certified for industrial deployment
- ISO 26262 and IEC 61508 safety certifications require documented engineering processes and third-party sign-offs the open-source community cannot provide
- ROS2's transport layer is fragmented across three DDS implementations (Fast DDS, Cyclone DDS, RTI Connext). All three interoperate without security enabled, but enabling security breaks cross-vendor communication, meaning a mixed-vendor fleet cannot be simultaneously secured and interoperable

## History and Trajectory of Middleware Development

**Era 1: ROS 1 (2007-2017):** Academic open-source project that prioritized performance over security, network topology, and system uptime

**Arise of the Commercial Gap:** ROS1's lack of security (authentication/ access control) and absence of real-time made it unsuitable for industrial deployment, exposing the need for a production-grade middleware layer

**Era 2: ROS2 (2017–present):** Built on the DDS communication standard, addressing ROS1's limitations, but inherited the same nonprofit governance model, leaving production scale challenges unsolved

**Era 3: Commercial Middleware Layer (2020 – present):** Independent companies emerged to solve production-scale gaps in ROS2. This includes Apex.AI, Foxglove, PickNick, and Open Navigation

**The consolidation Era:** The middleware converges around two competing outcomes:

1. Independent companies solving problems that require specialization or production-scale expertise will achieve a defensible position that full-stack platforms cannot replicate
2. Full-stack dominance. NVIDIA Isaac and Google's Intrinsic are two products with the potential to become a de facto standard across simulation, models, and runtime

Sources: [1] Aartsen et al., IEEE MED 2022, [2] Apex.AI ISO 26262, [3] CNBC, [4] Intrinsic, [5] Macenski et al., Science Robotics, [6] Mordor Intelligence, [7] NVIDIA Isaac, [8] NVIDIA Press Release, [9] Vaia

# THE APPLICATION LAYER

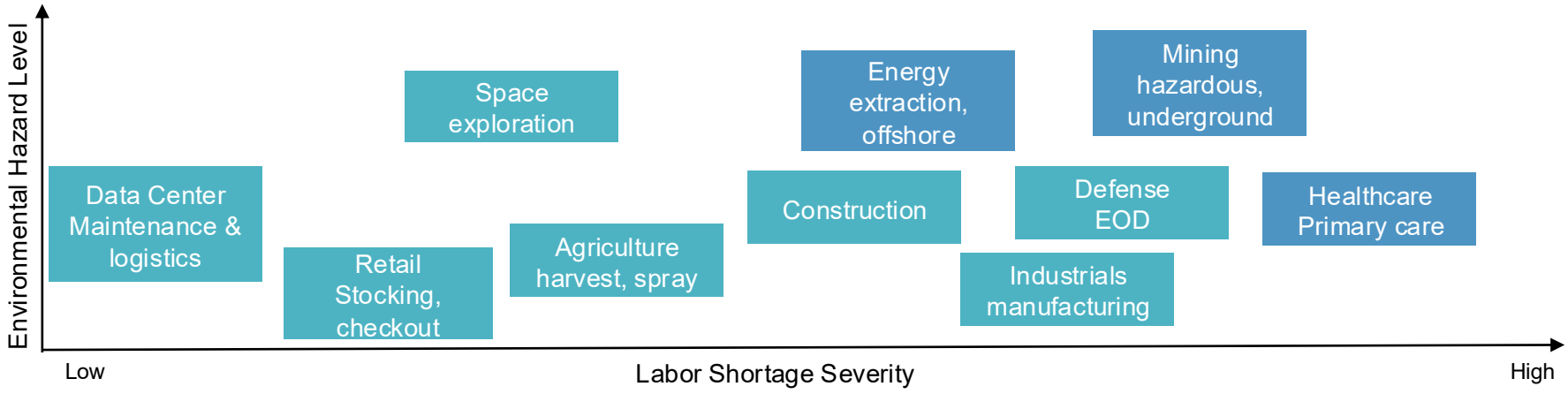
Field Service Robotics and Primary Care Robotics stand out as key investment potentials with macroeconomic labor shortages, geopolitical risks, and greater physical capacity and efficiency.

## Field Service Robotics

- Energy and critical minerals are bottlenecks of the AI boom and accelerating geopolitical risks, demanding higher output from existing sites rather than opening new ones. Yet current infrastructure is aging and industry is consistently ranked with the highest fatality rate, labour shortage compounds risks
- Autonomous systems can inspect, maintain, and repair aging pipelines and grid infrastructure continuously, catching failures before they cascade. Reliable autonomy in GPS-denied environment prevent exposing workers to hazards
- Hardware costs are no longer the challenge. Force-torque sensors fell from over $1,000 to under $10 per unit, and the tech stack followed similar price deflations across the board

## Primary Care Robotics

- By 2030, one in six people will be aged 60 or over, driving demand for diagnostics, early detection, and chronic disease management. Meanwhile there's a shortage of 70,610 primary care physicians by 2038, a job with low pay and high stress
- Now robotics can absorb the high-frequency, low-complexity work like preventive screenings, checkups, and monitoring, patient tolerance is increasing for robotic-assisted care
- The market remains a relative white space despite its eventual scale. Assistive health robotics growing at 24.6% CAGR and elderly care at 15.5% signal the demand and technical foundations. Regulatory frictions are keeping the first-mover window open for startups willing to navigate them




Sources: [1] Berkeley [2] Berkeley [3] Clayton Johnson [4] Deloitte [5] Epoch [6] Futurity [7] IQT [8] Substack [9] Opagio

# MOAT ANALYSIS

Robotic moats vary significantly in durability by layer, with open-source commoditizing generic capability while simultaneously creating the foundation for verticalized advantages

## Analysis Framework

### 1. Type of Moat

**Switching Costs**
Must align with existing infrastructure, combining specialized machines with software and long-term reconfiguration

**Pricing Power**
Often due to scarce data, safety certification, verticalization, or fleet-scaled pricing; Highly sub-sector dependent

**Open-Source**
Open-source can be both an erosive force (commoditizing generic capability) and a strategic moat

### 2. Current State of Moat

**Compounding**
Moat deepens with usage and strengthens over time which can increase exit costs non-linearly

**Static**
Durable but not growing, often structural such as safety certifications, contractual lock-in, IPs

**Eroding**
Actively shrinking from open-source, geopolitical tensions, commoditization, etc.

### 3. Impacts of Open-Source

**Hardware**
Minimal direct impact on physical manufacturing, open-source can't replicate proprietary methods or IP. Primary pressure is at the hardware-adjacent software layer

**Data**
Eroding at simulation and synthetic data (NVIDIA Isaac Sim). More slowly for high-fidelity real-world operational data which remains structurally scarce and can't be replicated

**Middleware/OS**
Most open source-saturated layer in the stack, more than 60% of robotic development strategies centered around ROS

**Models**
Open-source is closing the gap on generic models. OpenVLA outperforms RT-2-X by 16.5% on task success across 29 tasks

- **Hardware:** High, asymmetric, compounding switching costs, but pricing power is eroding in the mid-market as Chinese OEMs close the quality gap and software intelligence decouples value from hardware
  - Full lifecycle cost runs 3-5x the robot's purchase price, integration alone accounts for 30 to 50% of total project cost
- **Data:** Real-world data is irreplaceable because it captures contact dynamics, sensor noise, and rare events that simulators miss
  - Value exists in high-precision expert data, long-horizon mobile manipulation, humanoid embodiments
- **Foundation Models:** Switching cost of fine-tuning data and task-specific training is the most durable
  - However, open-world generalization is reducing the advantage of proprietary solutions
- **Middleware/OS:** Safety certification acts as the deepest switching cost, built against a specific middleware configuration
  - Despite open-source pressure, differentiation is achieved through safety, verticalization, or enterprise support SLAs

Sources: [1] Agility Robotics [2] Aitronik [3] AMD Machines [4] ArXiv [5] Business Wire [6] Codatta [7] Global X Label Studio [8] ReAnIn [9] Standard Bots [10] Tectonic [11] There's A Robot For That [12] V7 Labs

# MOAT ANALYSIS

Robotics is in its compounding phase with prioritization in full-stack breadth, but defensibility concentrates in the data and middleware layers which generate the strongest self-reinforcing advantages at deployment scale

- Proprietary data advantage belongs exclusively to companies generating irreplaceable real-world operational data as a byproduct of commercial fleet deployment
  - The data layer alone is structurally vulnerable; commodity-level robotics data (basic manipulation tasks, pick-and-place, structured environment navigation) is available through open-source datasets such as Open X-Embodiment, RoboSet, DROID, and RH20T
  - Mind Robotics's early partnership with Rivian supplies real-world production data from EV manufacturing in building AI-powered robots to focus on technical execution
- Vertical AI models or agents built on top of generalist foundation or world models are necessary to process specific contextual and multimodal (visual, spatial, temporal, etc.) robotics data
  - Unlike LLMs, "generalist models may get you 80% of the way there, robotics may start closer to 60%" thus requiring verticalized tuning and implementations
  - Fine-tuning is resource-intensive and an eroding moat, studies on "Re-Emergent Misalignment" show instead of adding new skills, narrow fine-tuning can erode previous knowledge and safety constraints
  - Durability in verticalized middleware is created because collaborative robots operating near humans must comply with safety standards like ISO 10218 and ISO/TS 15066
- Ultimately, integrating deployment infrastructure with layers combining models, middleware, proprietary data can create a self-reinforcing proprietary loop coined as "Innovation Stacks"
  - According to the "Robot-Powered Data Flywheel" framework, foundation models struggle in unstructured, real-world settings because deployment-time data (occluded objects, multilingual text, environmental noise) are greatly underrepresented in training sets
  - Models degrade when deployed unless continuously retrained on real data, VLA-based manipulation system in a warehouse distribution shift from lab to warehouse degrades performance from 95% to 80%

Sources: [1] ArXiv [2] ArXiv [3] a16z [4] Codatta [5] Insight [6] Ketryx [7] Mind Robotics [8] VC Cafe

# BOTTLENECK ANALYSIS

The absence of production-certifiable runtime at the system layer, where hardware and middleware meet, prevents deterministic integration, safety validation, and the ability to scale robot platforms




Deployments stall when robots must run deterministically and safely in brownfields or harsh environments. At the application layer control software must drive physical robots safely while staying tightly integrated with existing systems. Hardware interoperability issues affect 36% of manufacturers, pointing to failures at the middleware layer

## Overview

- Bottlenecks force teams to rebuild integration and safety cases for every robot and site
- Healthcare and space face added constraints, requiring regulatory approval (FDA, IEC 62304) and rigorous hardware validation (radiation, vibration, thermal testing) on top of core platform requirements
- Middleware constraint is a productization gap, not a pure research gap: ROS2, real-time Linux, and DDS can achieve near-deterministic lab performance, but are fragile to configure, debug, and certify at scale, so each team still rebuilds a bespoke stack for every robot and site

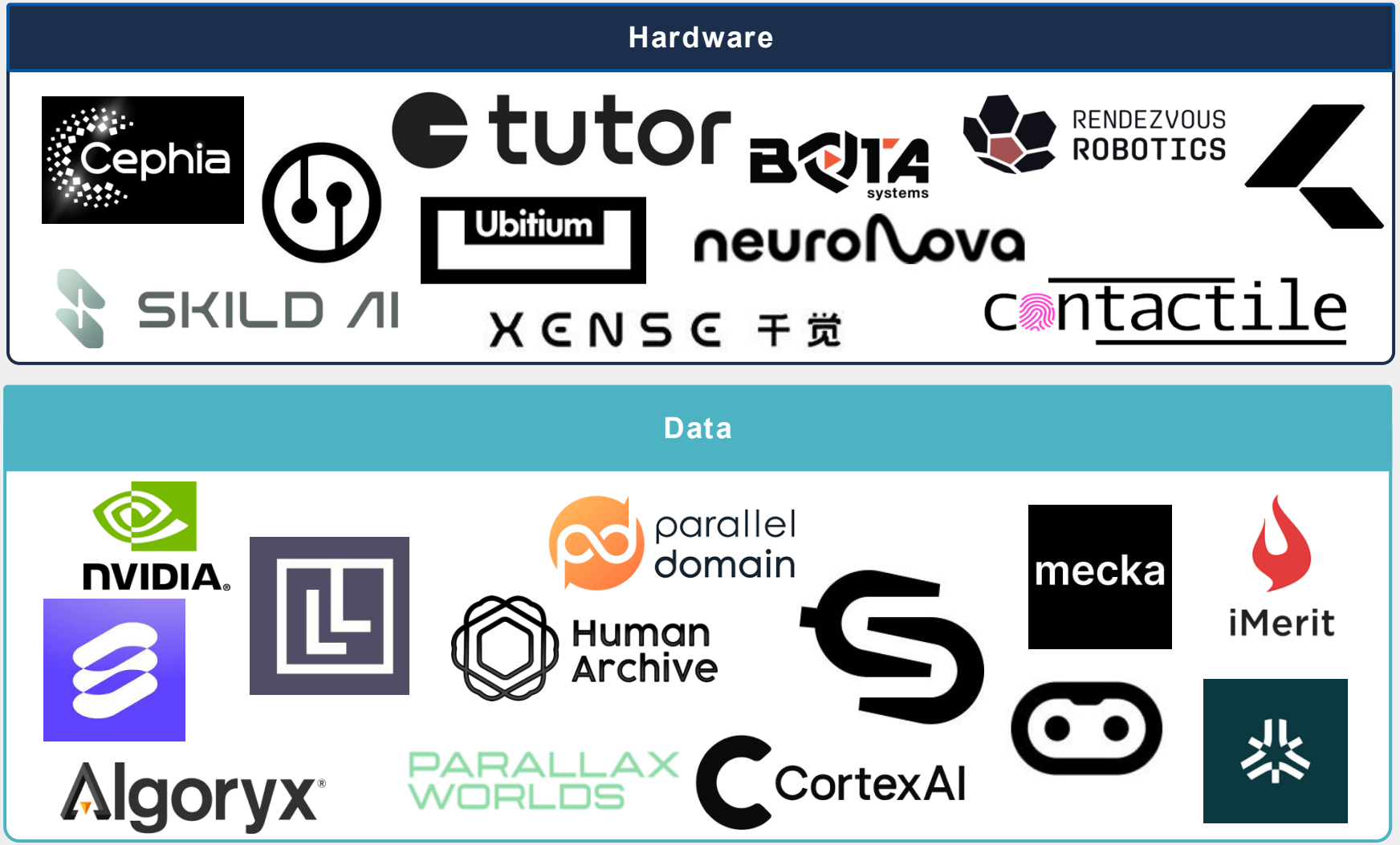
## Opportunities

- Middleware today has a relatively small, verticalized TAM and faces strategic pressure from hyperscalers and AI labs that are building pieces of the stack
- A durable business needs to own the safety-critical runtime and the surrounding compliance infrastructure, bundling execution, tooling, and reusable certification. Cloud providers focus up-stack and leave the middleware market underserved
- Thus, the core missing piece is a “robot OS” that allows developers to deploy and scale robots without customizing design and integration for each project

Sources: [1] a16 News [2] Eureka [3] Market Intel [4] Parasoft [5] Robotics Tomorrow [6] Ros Industrial [7] Sigbed [8] SQ Magazine

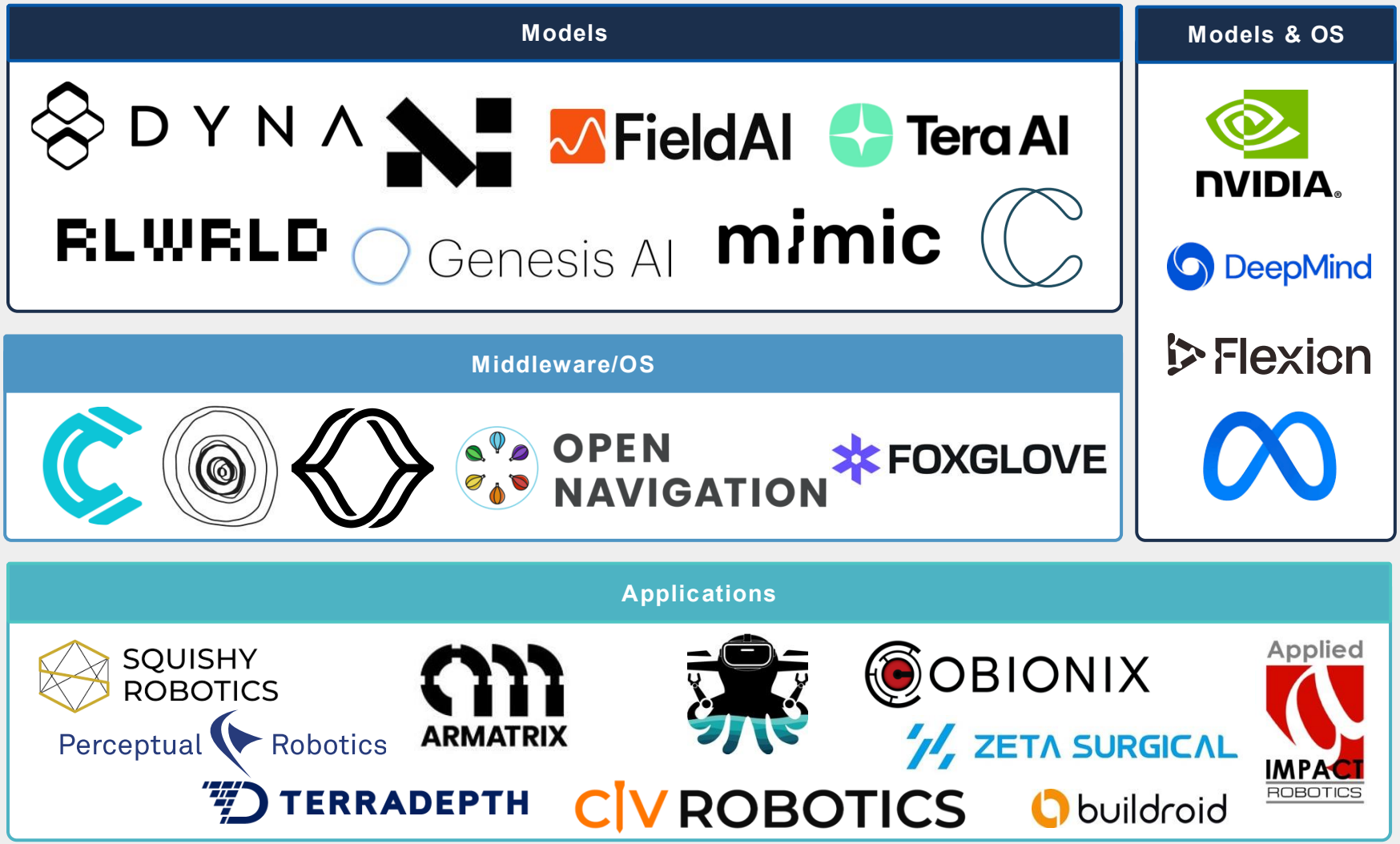
# MARKET MAP (I)

Density is highest in data and models, but middleware and hardware remain underserved relative to their role in production deployment, with middleware the clearest bottleneck and the fewest startups building against it



# MARKET MAP (II)

Density is highest in data and models, but middleware and hardware remain underserved relative to their role in production deployment, with middleware the clearest bottleneck and the fewest startups building against it



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# Questions?